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QUESTION 4 • UNIT 2 – LOSS FUNCTIONS

What is the Loss function? Enlist and Explain different Loss functions used in Neural Network.

📝 5 Marks

Loss Functions in Neural Networks

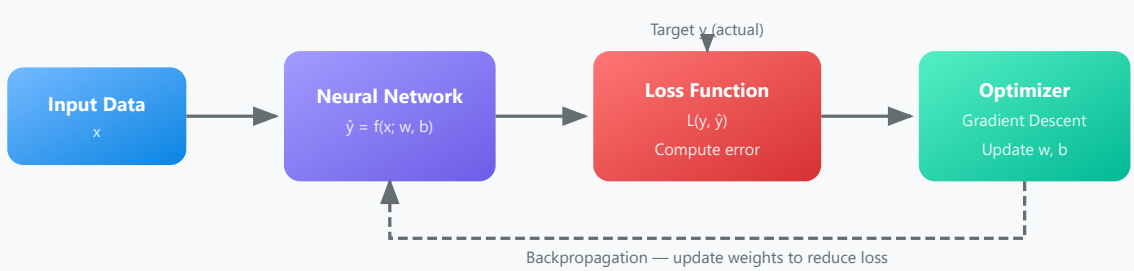
Definition, types with formulas and visual curves

What is a Loss Function?

A loss function (also called cost function or objective function) is a mathematical function that measures the difference between the predicted output of a neural network and the actual target value. It quantifies the error or deviation of the model's predictions from the true values.

The goal of training a neural network is to **minimize this loss**, thereby improving the accuracy of the model. Loss functions guide the optimization process during training — they help the model learn by providing feedback on how well it is performing and how it can adjust its parameters (weights and biases) to reduce the error.

Figure 1 — Loss Function in the Neural Network Training Loop



The loss function sits at the heart of the training loop — it measures prediction error and drives the optimizer to improve the model.

Types of Loss Functions

Figure 2 — Regression vs. Classification Loss Functions

Regression Losses
(continuous output)

- MSE
- MAE
- Huber Loss
- Log-Cosh

Classification Losses
(discrete output / probability)

- Cross-Entropy
- Hinge Loss
- KL Divergence
- Categorical CE

Loss functions are broadly divided into regression losses (for continuous targets) and classification losses (for discrete labels/probabilities).

1. Mean Squared Error (MSE)

$$\mathrm{MSE} = \frac{1}{N} \Sigma_{i=1}{}^{N} (y_i - \hat{y}_i)^2$$

Measures the **average squared difference** between actual and predicted values. Commonly used for **regression tasks**.

✓ Easy to interpret ✓ Penalizes larger errors more heavily

✗ Not robust to outliers ✗ Error unit is squared (not intuitive)

2. Mean Absolute Error (MAE)

$$\mathrm{MAE} = \frac{1}{N} \Sigma_{i=1}{}^{N} |y_i - \hat{y}_i|$$

Measures the **average absolute difference** between actual and predicted values. Also used for regression tasks.

✓ Robust to outliers ✓ More intuitive error measure

✗ Does not penalize large errors as strongly as MSE

3. Binary Cross-Entropy Loss

$$L = -\frac{1}{N} \Sigma_{i=1}{}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Used for **binary classification** tasks. Measures the difference between the predicted probability distribution and actual labels.

✓ Works well for probabilistic models ✓ Encourages predicted probability close to actual label

✗ Computationally expensive for large datasets

4. Hinge Loss

$$L = \Sigma_{i=1}{}^{N} \max(0, 1 - y_i \cdot \hat{y}_i)$$

Used for training classifiers, particularly **Support Vector Machines (SVMs)**. Penalizes predictions that are incorrect or not confident enough.

✓ Encourages correct classification with a margin

✗ Not suitable for probabilistic models

5. Huber Loss

$$L = \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2 & \text{if } |y_i - \hat{y}_i| \le \delta \\ \delta \cdot |y_i - \hat{y}_i| - \frac{1}{2}\delta^2 & \text{if } |y_i - \hat{y}_i| > \delta \end{cases}$$

Combines MSE and MAE. Behaves like **MSE for small errors** and **MAE for large errors**. Used for regression tasks.

✓ Robust to outliers ✓ Combines benefits of MSE and MAE

✗ Requires tuning of hyperparameter δ

6. KL Divergence Loss

$$D_{KL}(P \parallel Q) = \Sigma_i P(i) \log \frac{P(i)}{Q(i)}$$

Measures how one probability distribution Q differs from a reference distribution P. Used in **probabilistic models and generative models**.

✓ Useful for comparing probability distributions

✗ Sensitive to small probabilities

7. Log-Cosh Loss

$$L = \Sigma_{i=1}{}^{N} \log(\cosh(\hat{y}_i - y_i))$$

Similar to MSE but **less sensitive to outliers**. Used for regression tasks. Smooth and differentiable everywhere.

✓ Smooth and differentiable ✓ Robust to outliers

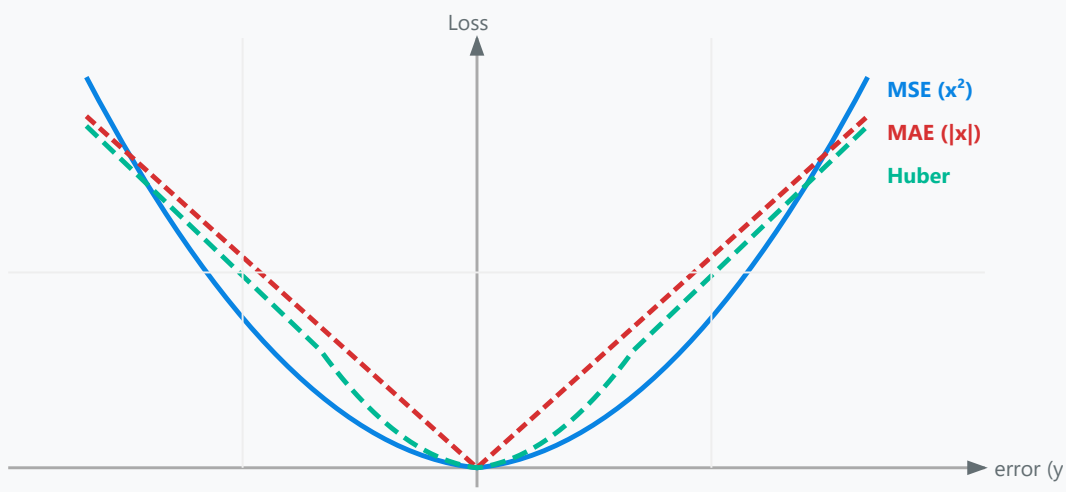
8. Categorical Cross-Entropy Loss

$$L = -\Sigma_{i=1}^{N} \Sigma_{j=1}^{K} y_{ij} \log(\hat{y}_{ij})$$

Used for **multi-class classification** problems. Measures the difference between the predicted probability distribution and the actual one-hot encoded labels.

Visual Comparison — MSE vs MAE vs Huber

Figure 3 — Loss Curves for MSE, MAE, and Huber



MSE penalizes large errors quadratically · MAE is linear · Huber combines both

MSE grows quadratically with error (sensitive to outliers), MAE grows linearly (robust), and Huber transitions smoothly between the two.

Importance of Loss Functions

✓ **Guides Learning** — helps the model understand how far its predictions are from the actual values.

- ✓ **Optimization** — used by optimization algorithms (e.g., gradient descent) to adjust model parameters.
- ✓ **Performance Metric** — indicates how well the model is performing during training.
- ✓ **Problem-Specific** — the choice of loss function depends on the type of problem (regression, classification, etc.) and the desired behaviour of the model.